MTH 203: Sequences and series

End-semester examination: 29/11/2025,
Time: 9:AM to noon

Total points: 35

Problem 1. (2 points each)

(a) State the Absolute Convergence Test for series.
(b) What is a Cauchy sequence?
(c) State the Nested Interval Property of the real numbers.

Problem 2. (2 points) Calculate $\lim_{n\to\infty}(n!/n^n)$.

Problem 3. (3 points) Calculate the radius of convergence of the power series $p(X) = \sum_{n=1}^{\infty} \frac{n^2}{4^n} X^n$.

Problem 4. (6 points) Let the sequence $(y_n)_{n\in\mathbb{N}}$ be defined inductively by setting $y_1 = 1$ and $y_{n+1} = 5 - 1/y_n$. Prove that $\lim_{n\to\infty} y_n$ converges and compute the limit. (Hint: Prove that the sequence is monotone.)

Problem 5. (3 points each) Classify the given series as divergent, conditionally convergent or absolutely convergent:

(a)

$$\sum_{n=1}^{\infty} \left(\frac{n^2}{n^3+1}\right)^n$$

(b)

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n + n^{3/2}}$$

Problem 6. Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be series of real numbers. Prove or disprove the following statements. (If the statement is not true, you should give a counter-example.)

(a) (3 points) If both the series are convergent, the series $\sum_{k=1}^{\infty} a_k b_k$ is also convergent.
(b) (3 points) If both the series are absolutely convergent, the series $\sum_{k=1}^{\infty} a_k b_k$ is absolutely convergent.

Problem 7. (6 points) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers such that the series $\sum_{n=1}^{\infty} |a_{n+1} - a_n|$ converges. Prove that the sequence $(a_n)_{n\in\mathbb{N}}$ converges. (Hint: You could try to prove that the sequence is a Cauchy sequence.)